CSC 163: Computing, Ethics, and Society

Computing and Society: Benefits, Harms, and Who Is Affected

ABET SO4, performance indicator 4d

Roadmap

1. **Professional responsibility:** the ACM Code
2. **A framework** for analyzing a computing solution
3. **Effects:** individuals, groups, society, environment
4. **Case studies:** facial recognition, ride-sharing, hiring algorithms, generative AI
5. **Mitigation** and trade-offs
6. **How to answer a 4d-style question**, with a model answer

Learning Objectives (SO4, PI 4d)

After this lecture you can:

- Identify the **stakeholders** of a computing solution
- Describe **positive and negative effects** on individuals, groups, society and the environment
- Use evidence (data, cases, standards) to support claims
- Weigh benefits against harms and propose **mitigations**
- Cite the **ACM Code of Ethics** to justify professional judgments

Professional Responsibility

Why engineers must think about society

The ACM Code of Ethics

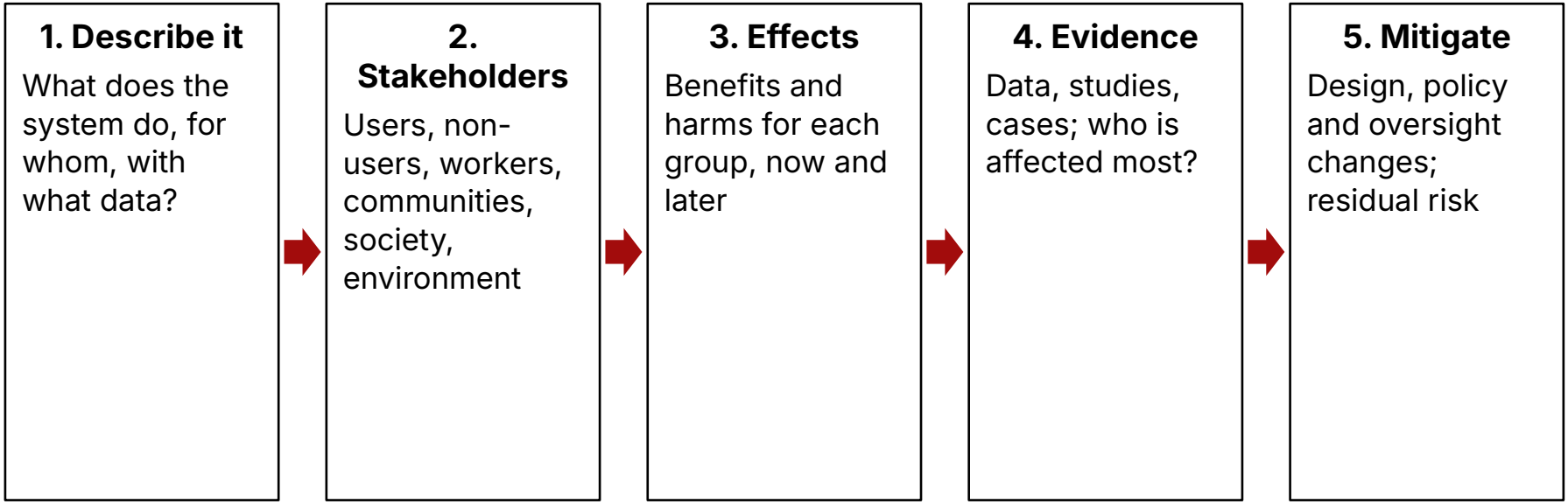
- **1.1 Contribute to society and to human well-being**, acknowledging that all people are stakeholders in computing
- **1.2 Avoid harm**
- **1.4 Be fair and take action not to discriminate**
- **1.6 Respect privacy**
- **2.9 Design and implement systems that are robustly and usably secure**
- The Code asks professionals to consider the **public good** in every decision

Source: ACM Code of Ethics and Professional Conduct (2018). Section numbers as published by ACM.

An Analysis Framework

Stakeholders, effects, evidence

Analyzing a Computing Solution



Always look for people who use the system **and** people who are affected without choosing to be.

Kinds of Effects

Level	Positive example	Negative example
Individuals	Faster service, convenience, access	Privacy loss, lost job, wrongful decision
Groups	Services for under-served communities	Unequal error rates; discrimination
Society	Public safety, economic growth, knowledge	Surveillance, misinformation, concentration of power
Environment	Efficiency; smarter energy use	Energy and water for data centers; e-waste
Time	Long-term productivity gains	Dependence, skill loss, effects that appear years later

Also ask: are the effects intended or unintended, and reversible or permanent?

Case Studies

Evidence-based analysis

Case 1: Facial Recognition

- **Benefits:** unlocking phones, finding missing people, speeding border checks, investigative leads
- **Harms:** mass surveillance; wrongful arrests; chilling effect on public life; biometric data cannot be changed after a breach
- **Evidence:** NIST's Face Recognition Vendor Test (Part 3, demographic effects) tested 189 algorithms from 99 developers on about 18 million images. Most algorithms had **false positive rates that differed by demographic group**, by factors from about 10 to more than 100 depending on the algorithm and the group
- **Meaning:** the same system helps some people and harms others, so accuracy must be measured **by group**

Source: NIST IR 8280 (2019)

Reading a False Positive Disparity

	Group A	Group B
False positive rate (FPR)	0.1%	1.0%
Comparisons per day	10,000	10,000
Expected false matches per day	10	100
Disparity ratio (B / A)		**10 times**

Same system, same threshold: members of group B are 10 times as likely to be wrongly matched. In a watchlist setting, that means more wrongful stops.

Case 2: Ride-Sharing Platforms

Benefits

- Convenient, affordable rides
- Flexible income for drivers
- Mobility where taxis are scarce

Harms

- Driver pay and benefits disputes over worker classification
- Congestion in cities
- Pressure on taxi drivers

Questions

- Who bears risk: the platform or the driver?
- Who is excluded: no smartphone, no bank card?
- How is data on riders and drivers used?

Different stakeholders experience the same system very differently.

Case 3: Automated Hiring Tools

- **Promise:** screen thousands of applications quickly and consistently
- **Amazon (reported by Reuters, 2018):** an experimental tool trained on past resumes learned to **downgrade resumes** that included the word 'women's'. Amazon stopped using it
- **Why it happened:** the training data reflected past hiring patterns, so the model copied the bias
- **Lessons:** historical data can encode discrimination; test outcomes across groups; keep **human review**; ask whether the tool is needed at all

Source: Reuters, Amazon scraps secret AI recruiting tool that showed bias against women (2018)

Case 4: Generative AI

Benefits

- Drafting, coding help, translation
- Accessibility tools
- Faster prototyping and learning

Harms

- Convincing misinformation and deepfakes
- Unclear copyright of training data
- Errors presented with confidence
- Job disruption

Mitigations

- Disclosure and labeling
- Human verification of outputs
- Licensing and opt-out rules
- Testing for bias and safety

Weighing and Mitigating

From analysis to a judgment

Comparison Matrix of the Cases

Case	Main benefit	Main harm	Most affected	Key mitigation
Facial recognition	Security, convenience	Surveillance, misidentification	Groups with higher error rates	Per-group testing; limits on use
Ride-sharing	Access and flexible work	Worker insecurity, congestion	Drivers, city residents	Fair pay rules; transparency
Hiring tools	Speed and consistency	Encoded bias	Applicants from affected groups	Bias audits; human review
Generative AI	Productivity	Misinformation, copyright, jobs	Creators, workers, public	Labeling; verification; licensing

Mitigation Strategies

- **Design:** privacy by design, diverse test data, safe defaults
- **Process:** impact assessments before launch, bias audits, red-teaming, user feedback channels
- **Oversight:** human review of high-stakes decisions, appeal paths, logging
- **Policy and standards:** regulation, industry standards, professional codes
- **Transparency:** tell people when a system is used and how it affects them
- Honest analysis names the **residual risk** that remains after mitigation

Answering a 4d Question

Structure, evidence, judgment

How to Answer a 4d-Style Question

1. Name the solution	Describe it in one or two sentences
2. Positive effects	Who benefits, how, with evidence
3. Negative effects	Who is harmed, how, with evidence; include non-users
4. Weigh and recommend	Balance, mitigations, ACM Code, residual risk

Level 4: several effects on several groups, supported by evidence, with a reasoned balance and mitigations.

Model Answer (Outline)

Question: Discuss the societal effects of using facial recognition for building access.

- **Solution:** cameras match faces to an employee database to open doors
- **Positive:** faster entry; no cards to lose; audit trail improves security
- **Negative:** biometric data is permanent, so a breach is serious; higher error rates for some groups lock employees out or let others in (NIST); employees feel watched
- **Judgment:** acceptable only with opt-out alternative, encrypted storage, retention limit, and per-group accuracy testing (ACM 1.2, 1.4, 1.6)

Summary

- Start with **stakeholders**, including people who did not choose the system
- Describe both **benefits and harms**, at individual, group, society and environment levels
- Back claims with **evidence**: NIST data, documented cases
- Weigh, **mitigate**, and name the residual risk; cite the **ACM Code**
- A strong answer is balanced, specific and evidence-based

Exercises and Solutions

Practice for performance indicator 4d

Exercise 1: Classify the effect

For each effect, mark it positive or negative, and the level it acts on: individual, group, society, or environment.

Effect	+ / –	Level
Face unlock saves a user time		
A higher false-match rate for one demographic group		
Constant public surveillance discourages protest		
Data centers consume large amounts of energy and water		
Telehealth gives rural communities access to care		
Deepfakes spread election misinformation		

Solution 1: Classify the effect

Name the level by who is affected: one person, a particular group, the public as a whole, or the physical environment.

Effect	+ / –	Level
Face unlock saves a user time	+	Individual
A higher false-match rate for one demographic group	–	Group
Constant public surveillance discourages protest	–	Society
Data centers consume large amounts of energy and water	–	Environment
Telehealth gives rural communities access to care	+	Group
Deepfakes spread election misinformation	–	Society

Exercise 2: Reading a disparity

A face-matching system runs 20,000 comparisons a day for each of three groups at the same threshold. Compute (a) expected false matches per day for each group and (b) each group's disparity ratio relative to group A. (c) Who needs attention first, and why?

Group	FPR	False matches / day	Ratio to A
A	0.05%		
B	0.5%		
C	1.0%		

Solution 2: Reading a disparity

Expected false matches = FPR × comparisons. Example: 0.05% × 20,000 = 10.

(c) Group C: its false matches are 20 times those of group A. Accuracy must be reported **by group**, because an overall average can hide this harm.

Group	FPR	False matches / day	Ratio to A
A	0.05%	**10**	**1×**
B	0.5%	**100**	**10×**
C	1.0%	**200**	**20×**

Exercise 3: Stakeholders of a hiring tool

A company adopts software that automatically screens resumes. For each stakeholder, name one possible benefit and one possible harm.

Stakeholder	Possible benefit	Possible harm
Applicants		
Groups under-represented in past data		
HR staff		
The company		
Society		

Solution 3: Stakeholders of a hiring tool

Include people who did not choose the system (applicants). Evidence: a reported 2018 case where a resume tool trained on past hiring learned to downgrade resumes containing the word 'women's'.

Stakeholder	Possible benefit	Possible harm
Applicants	**Consistent, quick response**	**Qualified people screened out unfairly**
Groups under-represented in past data	**Less individual manager bias, if designed well**	**Past bias reproduced by the model**
HR staff	**Less manual screening**	**Over-trust in the tool; less review**
The company	**Faster hiring, lower cost**	**Legal and reputation risk; missed talent**
Society	**More efficient labor market**	**Discrimination at scale**

Exercise 4: True or false

Write T or F. If false, correct it with a few words.

Statement	T / F	Correction (if false)
A system's effects fall only on its users		
High overall accuracy means no group is harmed		
Mitigation removes all risk		
A model trained on past hiring data can reproduce past bias		
Biometric data can be reset like a password		
A 4d answer should cover both benefits and harms		

Solution 4: True or false

1 point for T/F, 1 point for the correction.

Statement	T / F	Correction (if false)
A system's effects fall only on its users	F	Non-users and society are affected too
High overall accuracy means no group is harmed	F	Error rates can differ sharply by group; measure per group
Mitigation removes all risk	F	Residual risk remains and should be stated
A model trained on past hiring data can reproduce past bias	T	—
Biometric data can be reset like a password	F	It cannot be changed after a breach
A 4d answer should cover both benefits and harms	T	—

Exercise 5: Practice 4d question

Discuss the **societal effects** of a ride-sharing platform: who benefits, who is harmed, and what you would do about it. (Target: about half a page.)

Solution 5: Practice 4d question

- **Model answer.**
- Solution: an app matches riders with independent drivers.
- Positive: affordable, convenient rides; flexible income for drivers; mobility where taxis are scarce.
- Negative: disputes over driver pay and benefits (worker classification); added congestion; pressure on taxi drivers; riders' location data collected.
- Weigh and mitigate: clear pay and safety standards, limits on data use and retention, accessibility options; cite ACM 1.1 (benefit to society), 1.2 (avoid harm), 1.6 (privacy). Residual risk: platform power remains.
- **Scoring (4 points):** several effects on several groups (1); benefits and harms both covered (1); evidence or a concrete example (1); balance, mitigation, and a professional-ethics reference (1).

References

- ACM Code of Ethics and Professional Conduct (2018). acm.org/code-of-ethics
- NIST, Face Recognition Vendor Test Part 3: Demographic Effects, NISTIR 8280 (2019). nist.gov
- J. Dastin, Amazon scraps secret AI recruiting tool that showed bias against women, Reuters (2018)
- ABET, Criteria for Accrediting Computing Programs (student outcome 4). abet.org
- Original teaching summaries of public sources.